Supplementary Material for
Subset-Decomposed Evaluation of Schema-Aware Grounding for PostGIS Natural-Language-to-SQL Across Eleven LLMs

This supplement collects two pieces of material referenced from the main text but moved out of the main body to keep the manuscript focused. Cross-references from the main manuscript appear as "Supplement S1/S2".

- S1. Full baseline framework evaluation: BIRD warehouse track design-set and held-out, cross-lingual per-question re-audit, DIN-SQL external comparison, full reproducibility inventory, and the cross-model-family 2×2 factorial that motivated the broader cross-family panel reported in main-text Section 6.
- S2. End-to-end worked example tracing one Chongqing benchmark question through every pipeline stage (referenced from main text §3).

S1. Baseline framework evaluation (full version)

The main text §5 provides a compact summary of framework behaviour on the GIS Spatial set, BIRD warehouse track, DIN-SQL comparison, and cross-lingual probe; this section provides the full multi-section version with all per-question paired McNemar tables. These framework-level runs are supporting context and use their original protocols. The primary cross-family grounding-effect analysis in main-text Section 6 uses question-level majority vote followed by a single exact two-sided McNemar test on the per-question table.

S1.1 BIRD mini_dev design-set and held-out details

This section expands the warehouse-track stub in the main text §5. main text. We report the BIRD evaluation in two parts: a design set of 108 paired questions on which grounding rules were derived by error attribution, and a disjoint held-out set of 150 paired questions on which the same rules are tested without further tuning.

Round-1 grounding (design set). Round-1 adds three rule sections to the prompt: (i) multi-hop join-path discovery bridging fact-to-fact tables through pivot dimensions; (ii) aggregation semantics clarifying COUNT(*) vs. COUNT(col) vs. COUNT(DISTINCT col); (iii) temporal handling for TEXT-typed date columns (prefix comparison, SUBSTR extraction). On 101 paired questions it gave EX$= 0.564$ vs. baseline 0.515 (+0.050, paired McNemar two-sided $p = 0.2266$, $b = 3$, $c = 8$).

Round-2 grounding (design set). Error attribution on round-1 failures identified three patterns: 16 of 46 both-fails differed from the gold SQL only in DISTINCT placement (SELECT list dimensions duplicated through one-to-many JOINs); 4 regressions were caused by over-JOINing when a single table sufficed; multiple student_club questions used || concatenation where the gold returned first/last name as separate columns. Round-2 adds three targeted rules: DISTINCT-after-one-to-many-JOIN, avoid-over-JOIN, and output-column format (full name as separate columns; LIMIT 1 via ORDER BY rather than subquery with MAX/MIN).

Table 1 reports the round-2 result on 108 paired design questions and on the 150-question held-out set. Significance on the design set alone does not establish generalisation because the rules were derived from the same 108 items.

Table 1: BIRD warehouse results across rounds and splits.

Set	N	Baseline EX [95% CI]	Full EX [95% CI]	Δ [95% CI]
Round-1 grounding (design, subset)	101	0.515	0.564	+0.050 ($p = 0.227$, NS)
Round-2 design set (tuning)	108	0.500 [0.41, 0.59]	0.593 [0.50, 0.68]	+0.093 [0.02, 0.16] ($p = 0.021$)
Round-2 held-out (primary)	150	0.473 [0.40, 0.55]	0.507 [0.43, 0.59]	+0.033 [−0.03, 0.09] ($p = 0.383$, NS)
Validity (R2 design)	108	0.981	0.991	+0.010
Validity (R2 held-out)	150	0.980	1.000	+0.020
Mean tokens (R2 design)	108	1,010	7,975	7.9×

Interpretation. The round-2 design-set gain of +0.093 with $p = 0.0213$ should be read as a development-set tuning effect: the rules were derived by error attribution on this same 108-item sample, so significance on this set documents that the rules address the specific failure modes observed there. The primary warehouse-side significance test is therefore the held-out evaluation, where the same rules yield only +0.033 with 95% paired-proportion CI $[-0.03, 0.09]$ and $p = 0.3833$ on 21 discordant pairs ($b = 8$, $c = 13$); direction is preserved, magnitude drops by about two-thirds, and the CI comfortably covers zero. Two readings are consistent with this pattern: (a) part of the design-set gain is in-sample specialisation to the 108-item mix; (b) at $n = 150$ the effect-size-to-variance ratio remains low even if a small true effect exists. We therefore do not claim a BIRD-wide improvement from round-2 grounding. Validity improves on both sets (0.981 → 0.991 design, 0.980 → 1.000 held-out), which we attribute to the postprocessor's safety checks rather than to rule content. Token cost on the design set is 7.9× baseline, down from ∼32× in the earlier agent-loop configuration.

Representative design-set wins: Q1155 ("List patient ID, sex and birthday with LDH beyond normal range") — round-1 omitted DISTINCT, round-2 added it; Q1334 ("full names of members from Illinois") — round-1 used string concatenation, round-2 returned separate columns; Q1506 — round-1 missed DISTINCT, round-2 added it.

Relation to the earlier multi-pass evaluation. A multi-pass agent-loop run on a 500-question BIRD subset (without round-2 rules) previously reported baseline 0.474, full 0.501, $\Delta = +0.027$ with $p = 0.136$. We retain this only as context for the token-cost comparison; the 150-question held-out evaluation is the appropriate independent test of the single-pass grounded pipeline.

DIN-SQL older 495q comparison and held-out paired run. For the BIRD track, DIN-SQL was previously evaluated on a 495-question single-pass run as context for token cost; on that older sample, all three methods (Baseline 0.474, DIN-SQL 0.482, Full 0.501) cluster between 0.47 and 0.51 and the full-vs-DIN-SQL difference is +0.019 ($p = 0.382$, NS). The DIN-SQL row in Table 6 of the main text reports that older comparison only. We additionally run a paired DIN-SQL comparison on the round-2 150-question held-out set against the same Full pipeline configuration; the result is reported in Section S1.4 (Table 5): Full 0.507 vs. DIN-SQL 0.440, $b = 18$, $c = 8$, paired McNemar two-sided exact $p = 0.0755$ (marginal). The 495q row is retained for token-cost continuity; the held-out row is the appropriate independent DIN-SQL comparison.

S1.2 Cross-lingual re-audit: per-question detail

Table 2 lists the 50 BIRD Chinese items, their reviewer label (ok / fix / drop), the pre-review prediction correctness (0/1), and the post-review prediction correctness (0/1). This table is machine-generated from the re-evaluation harness.

Table 2: Per-question reviewer label and pre/post EX on the BIRD Chinese 50-question re-audit.

qid	db_id	difficulty	review	pre EX	post EX
1312	student_club	simple	fix	1	1
1317	student_club	moderate	fix	1	1
1322	student_club	moderate	fix	0	0
1323	student_club	moderate	fix	0	0
1331	student_club	simple	fix	0	0
1334	student_club	simple	ok	1	1
1338	student_club	moderate	fix	0	0
1339	student_club	challenging	fix	0	0
1340	student_club	moderate	fix	1	1
1344	student_club	simple	fix	1	1
1346	student_club	simple	ok	1	1
1350	student_club	moderate	fix	1	1
1351	student_club	simple	fix	1	1
1352	student_club	moderate	ok	1	1
1356	student_club	simple	fix	1	1
1357	student_club	simple	ok	1	1
1359	student_club	challenging	fix	1	1
1361	student_club	simple	ok	1	1
1362	student_club	simple	ok	1	1
1368	student_club	simple	ok	1	1
1471	debit_card_specializing	simple	ok	1	1
1472	debit_card_specializing	moderate	fix	1	1
1473	debit_card_specializing	moderate	ok	0	0
1476	debit_card_specializing	challenging	fix	1	1
1479	debit_card_specializing	moderate	fix	0	0
1480	debit_card_specializing	moderate	fix	1	1
1481	debit_card_specializing	challenging	fix	0	0
1482	debit_card_specializing	challenging	fix	0	0

continued on next page

Table 2 – continued

qid	db_id	difficulty	review	pre EX	post EX
1483	debit_card_specializing	simple	ok	1	1
1484	debit_card_specializing	simple	ok	0	0
1486	debit_card_specializing	simple	ok	0	0
1490	debit_card_specializing	moderate	fix	0	0
1493	debit_card_specializing	simple	fix	1	0
1498	debit_card_specializing	simple	fix	0	1
1500	debit_card_specializing	simple	fix	0	0
1501	debit_card_specializing	moderate	ok	0	0
1505	debit_card_specializing	simple	ok	0	0
1506	debit_card_specializing	moderate	fix	1	1
1507	debit_card_specializing	simple	fix	1	1
1509	debit_card_specializing	moderate	ok	1	1
1514	debit_card_specializing	simple	fix	0	0
1515	debit_card_specializing	simple	fix	1	1
1521	debit_card_specializing	moderate	ok	1	1
1524	debit_card_specializing	simple	fix	1	1
1525	debit_card_specializing	simple	ok	0	0
1526	debit_card_specializing	challenging	fix	0	0
1528	debit_card_specializing	simple	fix	1	0
1529	debit_card_specializing	moderate	fix	0	0
1531	debit_card_specializing	moderate	ok	0	0
1533	debit_card_specializing	moderate	fix	0	0

S1.3 Cross-model-family full factorial (Gemini vs. DeepSeek, 4 cells × N=3)

The initial framework evaluation used Gemini 2.5 Flash. To probe whether the observed gain from semantic grounding and intent-conditioned routing transferred to another LLM family, we ran a 2×2 factorial on the full 85-question GIS Spatial benchmark with model family ∈ {Gemini 2.5 Flash, DeepSeek-V3} and pipeline ∈ {schema-only baseline, full}. The full pipeline is routed through a unified model gateway so that the same ADK LlmAgent agent loop runs on either family without branch-specific code; DeepSeek is reached via the OpenAI-compatible LiteLLM adapter. Following the main-text Spatial protocol (Section 3.4 of the manuscript), the full cell is sampled N=3 times per family and aggregated by majority vote to absorb decoding stochasticity; each family uses its own provider-default temperature, which is deployment-realistic. DeepSeek baseline is also run at N=3 because DeepSeek's API is not deterministic at temperature 0; the Gemini baseline remains N=1 because the schema-only baseline_generate path sets temperature 0 explicitly and Gemini 2.5 Flash is deterministic for that family. Table 3 summarises the four cells; Table 4 reports the four paired McNemar tests on majority-vote predictions.

Reading. Three findings emerge. First, the cross-family schema-only baseline is exactly matched: both families reach 0.529 EX with zero discordant qids ($b = c = 0$, $p = 1.000$), so the full-pipeline contrast is not confounded by an a-priori capability gap between the two families on this benchmark. Second, the within-family Gemini grounding gain of +0.129 (majority-vote, $p = 0.052$) is consistent with the framework-level Spatial result. Third, the same full-pipeline stack — identical grounding rules, intent routing, self-correction, and postprocessor — does not transfer to DeepSeek: the

Table 3: 4-cell cross-family factorial on the full 85-question GIS Spatial benchmark. EX is execution accuracy over the paired gold-SQL harness; SD is sample SD across $N = 3$ samples within cell. Gemini full per-sample EX is reused from the main-text Spatial $N = 3$ protocol; DeepSeek cells are new.

Cell	EX mean	EX SD	N	Majority-vote EX
Gemini baseline	0.529	—	1	0.529
Gemini full	0.663	0.048	3	0.659
DeepSeek baseline	0.533	0.007	3	0.529
DeepSeek full	0.459	0.042	3	0.482

Table 4: Paired McNemar on majority-vote predictions across the 2×2 design (two-sided exact binomial). b is the number of qids where the first cell is correct and the second is incorrect; c is the converse.

Contrast	b	c	ΔEX	Paired p
Within-family Gemini (base vs. full)	8	19	+0.129	0.052
Within-family DeepSeek (base vs. full)	18	14	−0.047	0.597
Cross-family baseline (Gemini vs. DS)	0	0	0.000	1.000
Cross-family full (Gemini vs. DS)	22	7	+0.176	0.008

within-family DeepSeek contrast is in the opposite direction (-0.047, $p = 0.597$, null), and the cross-family full contrast is significantly in Gemini's favour ($+0.176$, $p = 0.008$). The grounding stack was authored and error-attributed against Gemini traces, and we interpret the DeepSeek regression as evidence that semantic grounding plus intent-conditioned routing, as currently instantiated, carries family-specific prompt sensitivities. This is an honest negative result for the portability claim: the mechanism works on the family it was tuned for, does not work on the unseen family at the same prompt budget, and the next question is whether family-adapted grounding rules recover the gain.

Protocol details. Full-pipeline per-sample EX values were: Gemini $\{0.706, 0.612, 0.671\}$; DeepSeek $\{0.494, 0.412, 0.471\}$. DeepSeek baseline per-sample EX: $\{0.529, 0.541, 0.529\}$. Temperature pinning was intentionally not applied across families: pinning temperature 0 in a cross-family study would sample a deployment configuration neither family actually ships; pinning temperature 1 would understate Gemini's deployed determinism. We therefore let each family run at its provider default and absorbed the decoding variance with $N = 3$ sampling per cell, which is the same protocol the main text uses for its Gemini Spatial headline. All six DeepSeek runs were executed on the same day, on the same benchmark snapshot, against the same PostGIS instance and the same gold-SQL harness.

S1.4 DIN-SQL held-out run (150q) and Robustness paired comparison

DIN-SQL is a published external baseline for text-to-SQL with explicit prompting stages (schema linking, classification-and-decomposition, self-correction). We report two paired comparisons of DIN-SQL against the proposed Full pipeline run with the same external baseline implementation, the same Gemini 2.5 Flash backbone, and the same execution-comparison harness, against PostgreSQL (BIRD) and PostGIS (Chongqing) targets.

BIRD 150q held-out. Table 5 pairs DIN-SQL with the Full pipeline on the 150-question BIRD mini_dev held-out set — the same held-out qid list used by the warehouse evaluation in Section 4.4 of the main text. The Full pipeline outperforms DIN-SQL by +0.067 EX, with discordant counts $b = 18$ in favour of the Full pipeline and $c = 8$ in favour of DIN-SQL; paired McNemar two-sided exact $p = 0.0755$ is marginal at $\alpha = 0.05$ and consistent with the small directional effect size on warehouse-style queries reported in Section 4.4. By BIRD difficulty, DIN-SQL reaches 0.585 (24/41) on simple, 0.434 (33/76) on moderate, and 0.273 (9/33) on challenging items, so the DIN-SQL deficit relative to the Full pipeline widens with question difficulty.

Table 5: Paired DIN-SQL vs. Full pipeline on the 150-question BIRD held-out set. Discordant counts use the convention b = Full-correct, DIN-incorrect; c = DIN-correct, Full-incorrect.

System	EX	Discordant b	Discordant c	Paired p
DIN-SQL	0.440 (66/150)	—	—	—
Full pipeline	0.507 (76/150)	18	8	0.0755

Robustness 40q. DIN-SQL was not designed for safety- or refusal-style queries; this is a stringency test of an external baseline against the Robustness suite, not a peer benchmark of two safety-aware systems. Table 6 pairs DIN-SQL with the Full pipeline on the 40-question Robustness suite. The Full pipeline outperforms DIN-SQL by +0.700 EX, with discordant counts $b = 28$ in favour of the Full pipeline and $c = 0$ in favour of DIN-SQL; paired McNemar two-sided exact $p = 7.45 \times 10^{-9}$. Per-category DIN-SQL accuracy: Security Rejection 3/11, Anti-Illusion 1/9, Schema Enforcement 2/3, Schema Hallucination 3/8, Data Tampering Prevention 0/1, OOM Prevention 2/8. The gaps on Security Rejection and Anti-Illusion in particular reflect the absence of a safety/refusal layer in the DIN-SQL prompt design.

Table 6: Paired DIN-SQL vs. Full pipeline on the 40-question Robustness suite. Discordant counts use the convention b = Full-correct, DIN-incorrect; c = DIN-correct, Full-incorrect.

System	EX	Discordant b	Discordant c	Paired p
DIN-SQL	0.275 (11/40)	—	—	—
Full pipeline	0.975 (39/40)	28	0	7.45×10^{-9}

S1.5 Full reproducibility inventory

Released artefacts. The anonymised companion repository includes the 125-question GIS benchmark with gold SQL (85 Spatial + 40 Robustness), the 108-question BIRD design set and 150-question held-out qid lists with full baseline and full-pipeline predictions, the 100-question cross-lingual pairs (50 Chinese BIRD + 50 Chinese GIS), the DIN-SQL runner adapted for PostgreSQL/PostGIS, the evaluation harness with McNemar and Wilson CI tools, the balanced N=5 gemini-3.5-flash A/B/C summary, the six-way ablation runner, and the SQL postprocessor and self-correction logic. Per-question predicted SQL and execution results are included. Revision source tables additionally include the fixed-host Gemma4 scale sweep, the queried Ollama model metadata, and the FloodSQL-Bench smoke-audit summary.

Table 7: Experimental configurations and released artefacts.

Experiment	Model	Key parameters
GIS 125q (baseline + full)	Gemini 2.5 Flash	temp=0.0, 60s timeout
GIS 125q (DIN-SQL)	Gemini 2.5 Flash	temp=0.0, 4-stage pipeline
BIRD 108q design + 150q held-out	Gemini 2.5 Flash	temp=0.0, single-pass
BIRD (DIN-SQL, older 495q)	Gemini 2.5 Flash	temp=0.0, 4-stage pipeline
Cross-lingual 50q BIRD + 50q GIS	Gemini 2.5 Flash	LLM-translated Chinese
Gemini 3.5 A/B/C diagnostic	Gemini 3.5 Flash	A/B/C all N=5, question-level majority vote
Gemma4 fixed-host sweep	Gemma4 e2b/e4b/12b/26b/31b	Ollama 0.30.5, GGUF Q4_K_M, CQ-125
FloodSQL-Bench smoke audit	Gemini 2.5 Flash	stratified 150q, N=1, local PostGIS port

Run-time configuration. The framework-level S1 runs use Gemini 2.5 Flash with temperature= 0.0, default top_p, and default maximum output tokens. Each question has a 60-second execution timeout; on 429 RESOURCE_EXHAUSTED, the runner retries up to 3 times with 2-second initial delay. On no-SQL output, the question is recorded with gen_status = no_sql and EX= 0. All McNemar results in this paper are reported as two-sided exact binomial tests on discordant pairs; the one-sided variant is not used because we do not have a pre-registered directional hypothesis on BIRD.

The Gemma4 sweep is documented by frozen local summaries and host metadata rather than by a repeated-sample inferential protocol. The FloodSQL-Bench audit uses a local PostgreSQL/ PostGIS port of the benchmark and a single gemini-2.5-flash stratified smoke run; it is included only to document that the external harness was operational.

Reviewer access. A reviewer-accessible companion-repository link is provided through the editorial system for double-anonymised review. The primary GIS result covers the 125-question benchmark (85 Spatial + 40 Robustness); the BIRD evaluation is split into a 108-question design set (rule derivation) and a 150-question held-out set (independent significance).

S2. Worked example: end-to-end PostGIS grounding

The main text §3 (Worked example) defers the full per-stage trace to this supplement. Figure 1 traces a single Chongqing benchmark question through every pipeline stage. The question asks for the total length in kilometres of one-way roads; the gold SQL uses ST_Length over the geography cast to return length in metres and divides by 1 000. Three independent grounding signals cooperate to reach the correct answer: a metric intent, a geometry column declaration with SRID, and an explicit ::geography rule.

Three implementation artefacts support this flow. The semantic layer is a per-database YAML file declaring, for each table, its aliases (including Chinese translations), geometry columns with SRID and geometry type, and a list of rule identifiers that should be injected when the table is matched. The intent rules are a small set of classifier tags (attribute_filter, aggregation, spatial_join, spatial_measurement, knn, preview_listing) each associated with a minimal set of prompt-side rules (for example, spatial_measurement enables the ::geography rule and a unit-

Stage	Content
NL question (q)	Chinese original, transliterated: tǒngjì suǒyǒu dānhángdào de zǒngchángdù (gōnglǐ). English gloss: “Total length in kilometres of all one-way roads.”
Intent classifier	intent = spatial_measurement, metric = length, unit = km, language = zh.
Semantic resolution (Stage 1)	Matches cq_osm_roads_2021 as the roads table; resolves the Chinese alias for “one-way road” to the filter oneway = ‘yes’; identifies geometry column (MultiLineString, SRID 4326, WGS 84 lat/lon).
Grounded context (Stage 2)	Schema dump for cq_osm_roads_2021; two injected rules: (i) metric operators on lat/lon data require ::geography cast so that ST_Length returns metres; (ii) unit conversion to km is a divide by 1 000 performed after SUM. One spatial-measurement few-shot example with geography cast.
Initial SQL $s^{(0)}$	SELECT ROUND(SUM(ST_Length(geometry)) / 1000.0, 2) AS length_km FROM cq_osm_roads_2021 WHERE oneway = ’yes’;
Postprocessor + execution	Executes, but returns a value in degree units because ::geography was not applied; the postprocessor tags this as a “metric unit sanity” regression against the few-shot expectation and feeds the error message back to the grounded generator.
Corrected SQL $s^{(1)}$	SELECT ROUND((SUM(ST_Length(geometry::geography)) / 1000.0)::numeric, 2) AS length_km FROM cq_osm_roads_2021 WHERE oneway = ’yes’;
Gold SQL	SELECT ROUND((SUM(ST_Length(geometry::geography)) / 1000.0)::numeric, 2) FROM cq_osm_roads_2021 WHERE oneway = ’yes’;
Outcome	Execution-based match (EX = 1); the postprocessor additionally adds the numeric cast required by PostgreSQL’s ROUND(double precision, integer) resolution.

Figure 1: Worked example: a Chongqing benchmark question traced through the full pipeline. The baseline omits the ::geography cast; Stage 2 rule injection closes the gap.

conversion hint). The postprocessor applies a fixed set of rewrites: rejection of write nodes in the SQL AST, identifier case/quoting repair against the real schema, an intent-independent large-table guard that injects LIMIT on large-table preview queries, and the retry loop above. A minimal set of these artefacts, together with the few-shot retrieval top-k setting and the prompt templates used in this paper, is released in the anonymised companion repository.